

Math 2551 Worksheet: Potentials and Conservative Vector Fields

1. Show that the vector field $\mathbf{F} = 12xy\mathbf{i} + 6(x^2 + y^2)\mathbf{j}$ is conservative using the mixed partials test, then find a potential function f such that $\mathbf{F} = \nabla f$.
2. Find a potential function f for

$$\mathbf{F}(x, y, z) = 2xy\mathbf{i} + (x^2 - z^2)\mathbf{j} - 2yz\mathbf{k}.$$

Evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where C is any path from $(0, 0, 0)$ to $(1, 2, 3)$.

3. Let a, b, c, d, e be real numbers and

$$P(x, y, z) = 3x + 7y + 2z;$$

$$Q(x, y, z) = ax + by + 4z;$$

$$R(x, y, z) = cx + dy + ez.$$

For which values of the constants a, b, c, d, e is $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ a conservative vector field?

4. Find a potential function f for

$$\mathbf{F}(x, y, z) = \left\langle \frac{1}{y}, -\frac{x}{y^2}, 2z - 1 \right\rangle$$

and use it to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the curve $C : \mathbf{r}(t) = \langle \sqrt{t}, t + 1, t^2 \rangle, 0 \leq t \leq 1$.

5. Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$ where the curve C is the unit circle oriented counterclockwise.